

SAITEJ CHOWDARY POLAVARAPU

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PROFESSIONAL SUMMARY

Experienced Data Engineer skilled in designing and implementing data solutions across various industries. Demonstrates expertise in cloud technologies, big data frameworks, and advanced analytics to enhance business value. Notable achievements include optimizing PySpark applications for healthcare data processing and integrating Delta tables for improved data governance. Aims to leverage technical skills and collaborative approach to drive data engineering success in a dynamic environment.

CORE QUALIFICATIONS

- **Languages:** Python, SQL, Shell Scripting
- **Cloud Services:** AWS (S3, EC2, EMR, Redshift, Athena, IAM), MS Azure (DataBricks, ADF)
- **Databases:** MySQL, PostgreSQL, Snowflake, Oracle, DB2, Redshift, Hive, Azure SQL DW, DataBricks Database
- **Big Data Technologies:** Hadoop (HDFS, MapReduce, YARN), Apache Spark, Apache Kafka, Apache Hive, Sqoop
- **Data Quality and Governance:** Apache Atlas, Collibra, Informatica Data Quality, Talend Data Quality, Trifacta
- **CI/CD and Version Control:** Kafka, Jenkins, Git, GitLab, Bitbucket
- **Data Visualization and Reporting:** Tableau, Power BI, Looker, QlikView, Google Data Studio
- **Python Libraries:** NumPy, Pandas, Beautiful Soup, PySpark, SQLAlchemy, PyTest

EXPERIENCE

Blue Cross Blue Shield

Dec 2022 - Present

Data Engineer

TX

- Refined PySpark applications to boost performance, enabling the precise processing of over 10 million healthcare records daily. Accelerated claims and policyholder data pipelines, cutting execution time by 35% while ensuring compliance with industry standards.
- Revamped the data ingestion workflow for third-party healthcare datasets, ensuring regulatory compliance with HIPAA while processing over 500,000 records weekly and achieving a lead time reduction of three days in reporting.
- Directed ETL operations with DataStage to seamlessly integrate insurance claims, policy data, and medical records, improving cross-departmental data flows. Boosted data processing efficiency by 30%, ensuring accurate and timely information exchange across teams.
- Redesigned data pipelines by optimizing Spark workflows, enhancing patient records management. Eliminated bottlenecks to streamline integration, reducing average report generation time from 48 hours to 12 hours.
- Leveraged advanced analytics to assess health outcomes through predictive modeling with PyTorch, resulting in a 20% increase in claim processing speed and enhancing overall service delivery for healthcare providers.
- Executed strategies for managing Delta tables that elevated the integrity of patient data handling; reinforced comprehensive security practices under BCBS's regulations while decreasing threats of unauthorized access by 40%.
- Streamlined DataBricks for real-time claims adjudication, cutting insurance claim processing time by 45% while ensuring data consistency and complying with industry regulations.
- Scaled AWS infrastructure (EC2, S3, EMR, Redshift) to accommodate expanding healthcare data requirements, improving analytics capacities and storage flexibility.
- Designed and managed robust ETL pipelines with Spark and Hive, cutting patient reporting delays by 40 hours per week. Elevated data accuracy, leading to faster decision-making in healthcare services.
- Built an interactive Tableau dashboard tracking 15 critical healthcare metrics, empowering leadership to access real-time insights and make data-driven decisions swiftly in response to evolving patient needs.
- Deployed real-time data streams by integrating Kafka and Spark Streaming, streamlining patient encounter workflows, and reducing response times from 8 minutes to under 2 minutes, enabling faster clinical decision-making.
- Developed data models in Snowflake and Amazon Redshift, improving the efficiency of analytics queries by 35% and enabling faster decision-making for claims adjudication.
- Implemented enhanced logging mechanisms within PySpark jobs facilitating better error tracking and debugging capabilities while reducing turnaround time on troubleshooting tasks by up to four hours weekly.
- Created machine learning algorithms using Python to predict patient diagnoses based on historical claims data. These predictive models increased early intervention rates across departments, boosting overall patient care efficiency by at least 30%.
- Established robust data integration workflows with Jenkins and Git that boosted deployment efficiency by 25%, allowing for faster updates to production environments without compromising system stability.
- Conducted data quality checks using Talend and Informatica, ensuring 99.9% accuracy in reporting.

Epsilon

May 2022 - Dec 2022

Big Data Engineer

TX

- Formulated comprehensive training programs for 50+ employees on best practices in data handling; these initiatives raised awareness about security protocols and led to an increase of reported vulnerabilities by 20%.
- Conducted thorough assessments of multi-layer security architectures across big data environments, presenting findings that directly addressed the three biggest causes of data breaches, fostering enhanced protection measures within organizational frameworks.

- Composed thorough guides for critical security procedures that addressed every aspect of data protection; accelerated the onboarding timeline for newcomers, allowing them to meet compliance requirements 30% quicker.
- Evaluated and reinforced the security architecture of big data systems, enhancing security frameworks and reducing vulnerabilities by 40%.
- Generated innovative navigation pathways in internal verification processes that increased experience ratings of a targeted group of 50 valued users while adhering strictly to robust safety guidelines promoting total process visibility.
- Spearheaded the transition to real-time data processing frameworks like Kafka, achieving sub-second latency for high-velocity streams; directly contributed to faster decision-making on critical business insights while handling over 1M transactions daily.
- Built and maintained data lakes and warehouses using technologies like Amazon S3 and Snowflake, which provided structured and unstructured data storage for analytics, and improved the flexibility of data management processes.
- Constructed efficient batch-processing pipelines employing precision techniques tailored for immediate analytical needs, transforming idle querying periods into actionable intelligence outputs-most junior member involved during development stages driving project success.

KPMG

Aug 2020 - Jul 2021

Data Engineer

India

- Authored complex stored procedures, saving 3 hours per week on manual report generation while maintaining compliance and quality assurance.
- Crafted intricate SQL queries and stored procedures within Oracle databases, leading to a 50% acceleration in data retrieval processes for regulatory compliance reports while ensuring adherence to stringent industry standards.
- Employed Matplotlib to create detailed visualizations of compliance data trends and patterns, empowering regulatory teams to identify potential risks and maintain adherence to regulatory requirements across 50+ compliance areas.
- Automated the collection and aggregation of regulatory datasets using advanced AWS services such as AWS Glue, Lambda, Redshift, and S3, saving over 300 hours annually in compliance report generation while ensuring full adherence to regulatory requirements.
- Created fully automated pipelines employing Hadoop ecosystem components, processing over 5TB of dataset daily, allowing the team to respond swiftly to evolving regulations without manual intervention or delays.
- Conceptualized and deployed interactive dashboards and reports in Power BI and MS Excel, translating complex regulatory data into clear and actionable visual formats that facilitated informed decision-making by compliance officers and executive leadership.
- Engineered ETL processes using Talend and SSIS, ensuring consistent and accurate data flow for regulatory reporting across 15 different compliance reports, minimizing the risk of errors and maintaining compliance with regulatory standards.
- Coordinated cross-departmental efforts leading up to successful completion of eight major deliverables within strict deadlines as part of implementing robust data processes aimed at ensuring no lapses in compliance occurred during operations.
- Architected real-time data pipelines utilizing Kafka and Apache Flink, facilitating the immediate processing of high-velocity data streams which resulted in a 30% improvement in time-to-insight for critical decision-making.

KPMG

Feb 2020 - Jul 2020

Data Analyst

India

- Analyzed large datasets from the Fraud Detection System using Python, NumPy, and Pandas, extracting key insights that improved organizational decision-making and optimized fraud detection strategies.
- Engineered over 20 SQL queries on MySQL databases for data retrieval, manipulation, and transformation of fraud-related big data, facilitating comprehensive reports and intensive scrutinizes.
- Architected robust ETL processes utilizing Azure Data Factory, establishing seamless data integration into the organization's warehouse for the Fraud Detection System; revamped workflows resulted in a 50% decrease in processing time.
- Executed intricate operations in Azure Databricks for advanced analytics and machine learning within the Fraud Detection System, resulting in a 40% increase in detection accuracy of fraudulent transactions through refined data processing workflows.
- Designed a suite of interactive reporting tools including 10 analytics dashboards via Power BI which streamlined access to vital information about detected anomalies while allowing team members quick reference during investigations into suspicious activity.
- Collaborated with cross-functional teams in an Agile environment, participating in sprint planning, daily scrums, and retrospectives, ensuring successful completion of data analysis work in the Fraud Detection System.
- Devised structured version control processes within the Fraud Detection System's development cycle, optimizing task execution time by ensuring seamless collaboration with over 10 data analysts using advanced Git tools.

EDUCATION

University of Dayton

Aug 2021 - Dec 2022

Master of Science, Computer Science

dayton ohio

- **GPA:** 3.9

Gitam University

May 2017 - Jun 2021

Bachelor of Technology, Electronics and Communications Engineering

india

- **GPA:** 3.5